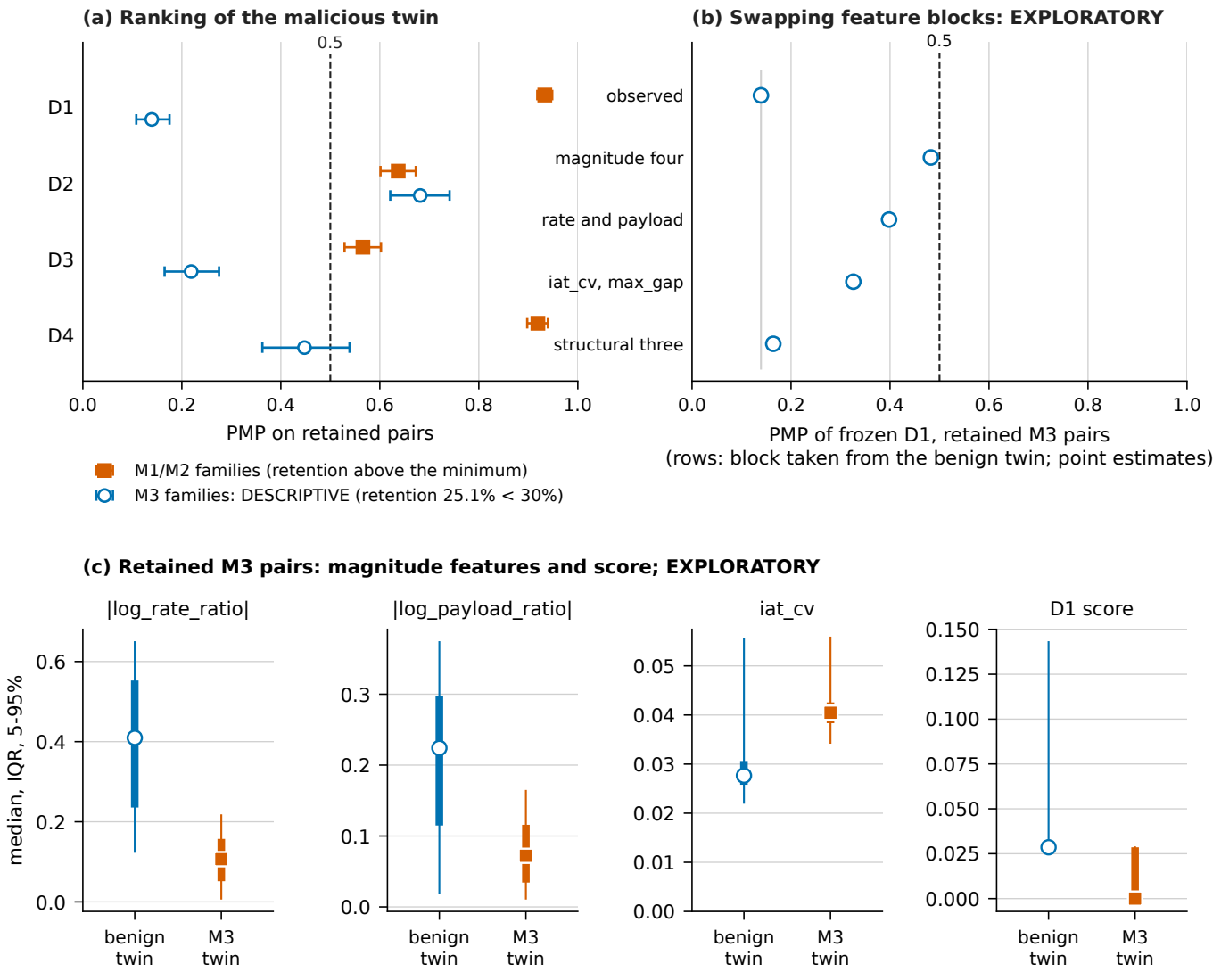
(a) Ranking of the malicious twin
0.5
D1
D2
D3
D4
0.0
0.2
0.4
0.6
0.8
1.0
PMP on retained pairs
M1/M2 families (retention above the minimum)
M3 families: DESCRIPTIVE (retention 25.1% < 30%)
(b) Swapping feature blocks: EXPLORATORY
0.5
observed
magnitude four
rate and payload
iat_cv, max_gap
structural three
0.0
0.2
0.4
0.6
0.8
1.0
PMP of frozen D1, retained M3 pairs
(rows: block taken from the benign twin; point estimates)
(c) Retained M3 pairs: magnitude features and score; EXPLORATORY
|log_rate_ratio|
|log_payload_ratio|
iat_cv
D1 score
median, IQR, 5-95%
0.0
0.2
0.4
0.6
0.1
0.3
0.00
0.01
0.02
0.03
0.04
0.05
0.000
0.025
0.050
0.075
0.100
0.125
0.150
benign twin
M3 twin